

Research Statement

Introduction

One of the most useful ideas in mathematics is *duality*: two kinds of information that appear different can determine one another. Just as the landscape shapes the trees that grow on it, the trees can also reveal the landscape beneath them. In algebraic topology, homology and cohomology provide two such perspectives on a space. My research studies how this principle appears in stable homotopy theory and arithmetic. The central goal of my research program is to understand duality and descent in *Real algebraic K-theory*, particularly how global information can be recovered from local data while retaining symmetry. My dissertation establishes a first step through a *K-theoretic duality theorem* for number fields. My current work extends this question to *Grothendieck–Witt theory* and develops *Real trace methods* for computation, while my work in equivariant algebra provides a foundation for organizing the symmetries and operations that arise in this broader program.

1 Background

Topology often turns geometric questions about spaces into algebraic ones. Homology studies geometric features inside a space, such as connected pieces, loops, and higher-dimensional holes. Cohomology begins from a different point of view: it studies algebraic data that behave like functions living on the space and how such data fit together globally. The *Universal Coefficient Theorem* explains that these two kinds of information are closely related. For example, with coefficients in a field k , this relationship takes the particularly simple form

$$H^n(X; k) \cong \text{Hom}_k(H_n(X; k), k),$$

The power of duality allows a problem expressed in one kind of information to be understood through a seemingly different kind.

The spaces and algebraic objects studied in topology often carry symmetries, which provide another important source of structure. For example, a circle can be reflected across an axis or rotated by half a turn. Reflection fixes two points, while the half-turn fixes none. Once the symmetries are forgotten, the underlying space is the same circle in both cases. Equivariant topology retains this information by studying not only the space itself, but also how its symmetries act and what remains fixed.

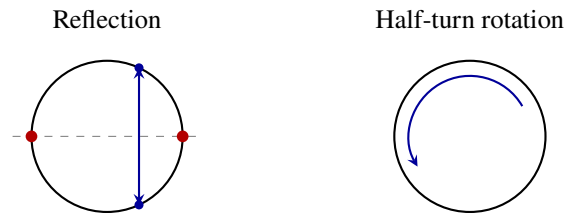


Figure 1: Two symmetries of the same underlying circle. Reflection fixes two red points, while rotation by half a turn fixes none.

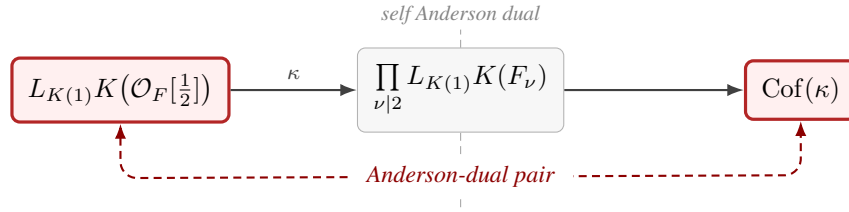
K-theory is a central setting in which both duality and equivariant phenomena arise. Topological K-theory studies vector bundles over spaces, while algebraic K-theory studies their algebraic counterparts, such as projective modules over rings. When the spaces, bundles, or rings under study carry symmetries, K-theory can be refined to retain them. Atiyah introduced Real K-theory to study complex vector bundles together with symmetry modeled on complex conjugation. Real algebraic K-theory is an algebraic variant of Atiyah's theory for rings with involution. It therefore provides a setting in which duality in algebraic K-theory can be studied while retaining symmetry.

2 Arithmetic Duality and Algebraic K-theory

2.1 K-theoretic Tate–Poitou duality. My dissertation asks whether a classical arithmetic duality has a spectrum-level form in algebraic K -theory. The starting point is Tate–Poitou duality, a central cohomological duality for number fields traditionally expressed through étale cohomology.

Although algebraic K -theory is notoriously difficult to compute, its $K(1)$ -localization provides a more tractable approximation that retains the arithmetic information relevant here. Through the Quillen–Lichtenbaum conjecture and the Thomason descent spectral sequence, this approximation is closely connected to étale cohomology [Tho85, RW00, Mit00]. This makes it natural to expect Tate–Poitou duality to lift to $K(1)$ -local algebraic K -theory and my dissertation confirms this expectation at the prime 2.

Theorem 1 ([Cho25]). *Let F be a number field. The completion cofiber sequence of $K(1)$ -local algebraic K -theory admits a canonical 2-adic Anderson self-duality. The duality acts by a vertical reflection, interchanging two end terms and identifies the middle term with its own dual.*



A homotopy-group-level form of this duality was conjectured by Calegari [CE12]. Blumberg and Mandell later formulated its spectrum-level refinement using Anderson duality $I_{\mathbb{Z}_p}$, and proved the corresponding local and global results at every odd prime [BM20]. The theorem above completes the spectrum-level picture at the remaining prime by accounting for the additional cohomological behavior arising from real embeddings.

At the heart of the proof is the Thomason descent spectral sequence, which connects $K(1)$ -local algebraic K -theory with étale cohomology. Through this bridge, Tate–Poitou duality appears as the cohomological shadow of Anderson duality, and the resulting spectrum-level equivalence recovers the classical Tate–Poitou exact sequence on homotopy groups.

$$\begin{array}{ccc}
 \text{Arithmetic duality:} & H_{\text{ét}}^*(\mathcal{O}_F[\frac{1}{p}]; \mathbb{Z}/p^k(*)) & \xleftarrow{\text{Pontryagin dual}} H_c^*(\mathcal{O}_F[\frac{1}{p}]; \mathbb{Z}/p^k(*)) \\
 \downarrow \text{Descent SS} & \Downarrow & \Downarrow \\
 \text{Homotopical duality:} & L_{K(1)}K(\mathcal{O}_F[\frac{1}{p}]) & \xleftarrow{\text{Anderson dual}} \text{Cof}(\kappa)
 \end{array}$$

Figure 2: The descent spectral sequence transfers arithmetic duality to homotopical duality.

The case $F = \mathbb{Q}$ also has an application to the algebraic K -theory of the sphere spectrum. Combining the duality theorem with known comparison results gives a canonical equivalence in nonnegative degrees

$$\Sigma^{-1}I_{\mathbb{Z}_2}L_{K(1)}K(\mathbb{Z}) \simeq_{[0, \infty)} \text{Fib}(\text{trcs} : K(\mathbb{S})_2^\wedge \longrightarrow TC(\mathbb{S})_2^\wedge). \quad (1)$$

Thus arithmetic information contained in the K -theory of the integers determines the error in a fundamental approximation to the algebraic K -theory of the sphere.

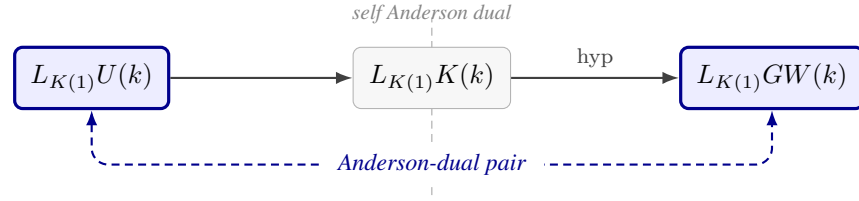
More broadly, this result provides a nonequivariant starting point for my study of duality in Real algebraic K -theory. It suggests that K -theoretic Tate–Poitou duality may be one shadow of a larger picture in which duality is compatible with involutions and C_2 -symmetry. My next projects approach this broader question through Grothendieck–Witt theory and Real trace methods.

2.2 Grothendieck–Witt theory and Real duality. Algebraic K-theory가 complex vector bundle의 analogue라면, Grothendieck-Witt theory는 real vector bundle의 analogue입니다. 이는, 대수적으로는 projective module 뿐만 아니라, 그 위의 nondegenerate bilinear form까지 보겠다는 말이기도 합니다.

ring에 involution이 있는 경우에, K-theory는 그 symmetry를 inherit하며, C2 fixed point가 바로 GW theory입니다.

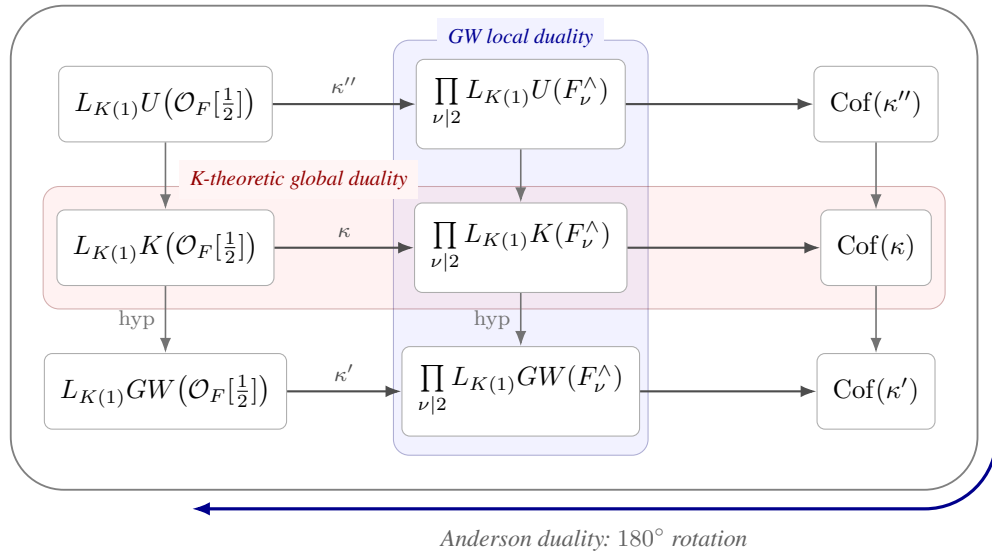
k 가 local field인 경우에, $K(1)$ -local GW 의 Anderson dual은 form twist를 2만큼 shift시킨다는게 conjecture.

Theorem 2 ([?]). *Let k be a 2-adic local field. The hyperbolic sequence of $K(1)$ -local Grothendieck-Witt theory admits a canonical 2-adic Anderson self-duality. The duality acts by a vertical reflection, interchanging two end terms and identifies the middle term with its own dual.*



Here, $U(k)$ is the homotopy fiber of the hyperbolic map.

Theorem 3 (Work in progress). *Let F be a number field. The completion cofiber sequences for U-theory, algebraic K-theory, and GW-theory assemble into a single diagram that is self-dual under 2-adic Anderson duality. The duality acts by a half-turn, exchanging the U- and GW-theoretic rows and carrying the K-theoretic row to itself. The middle row recovers K-theoretic Tate–Poitou duality, while the middle column recovers local GW duality.*



Thus, the K-theoretic Tate–Poitou duality in Theorem 1 appears as one slice of a larger duality pattern linking U-theory, algebraic K-theory, and Grothendieck–Witt theory.

Equivariant setting에서 Anderson duality; Ricka의 result. 위 결과를 real algebraic K-theory로 확장하는것.

2.3 Homological Real trace methods. My duality work studies structural relationships in Real algebraic K -theory; a complementary challenge is to develop tools for explicit computation. In joint work with Teena Gerhardt, Liam Keenan, Juan C. Moreno, and J.D. Quigley, I develop *homological Real trace methods* toward this goal [CGK⁺26]. Trace methods replace difficult K -theoretic questions by more computable invariants, but in the Real setting the additional C_2 -symmetry provides new structure that can itself be used in computation.

This extra symmetry already appears in Real topological Hochschild homology. While ordinary THH carries the familiar circle action, THR retains an additional reflection symmetry, as illustrated below.



Figure 3: The familiar $\mathbb{T}$ -action on THH extends in THR to an $O(2) \simeq \mathbb{T} \rtimes C_2$ -action.

Our homological Real trace methods are designed to exploit this additional genuine equivariant structure rather than forget it. A key feature of our construction is its compatibility with equivariant multiplicative and power operations, which we use to determine differentials and extract information about the continuous homology of TCR^- . Technically, this is realized by the following spectral sequence.

Theorem 4 ([CGK⁺26]). *For a C_2 -equivariant commutative ring spectrum A , there is a natural multiplicative spectral sequence*

$$H_\star(\mathrm{THR}(A); \mathbb{F}_2) \langle \bar{y} \rangle \implies H_\star^c(\mathrm{TCR}^-(A); \mathbb{F}_2),$$

compatible with equivariant power operations. For $MU_{\mathbb{R}}$, $BP_{\mathbb{R}}$, and several truncated Brown–Peterson theories, we determine the differentials and the E_∞ -page of the resulting spectral sequences.

These computations demonstrate the method on a fundamental family of spectra in equivariant chromatic homotopy theory. Our construction is a Real refinement of the homological trace methods of Bruner and Rognes [BR05], whose nonequivariant methods became important in the study of Segal conjectures for topological Hochschild homology of MU and BP . Motivated by this precedent, my long-term goal is to study a Real analogue of the Segal conjecture, beginning with the trace invariants of $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$. This would extend the present calculations toward a broader computational understanding of Real trace invariants and, ultimately, Real algebraic K -theory.

The role of genuine equivariant operations in these computations also raises a broader question: which collections of equivariant operations can arise, and how should they be organized? My work discussed next studies the algebraic and homotopical foundations of this question.

2.4 Equivariant algebra: operations and gradings. Symmetry changes not only the objects of stable homotopy theory, but also the algebra needed to describe their invariants. Genuine equivariant invariants carry richer operations and richer gradings than their nonequivariant counterparts. My work studies these two complementary aspects of equivariant algebra: which equivariant operations can coexist, and how representation-graded information should be organized.



Figure 4: Two complementary aspects of equivariant algebra

One part of this problem is to understand which equivariant operations can coexist. Transfer systems [Rub21] provide a combinatorial model for these choices, while N_∞ -operads encode the corresponding homotopical structures. In collaboration with David Chan, David Mehrle, Pablo Sanchez Ocal, Angélica Osorno, Ben Szczesny, and Paula Verdugo, we study whether compatibility at the combinatorial level is actually realized homotopically. Our main result establishes one direction in general and a converse in an important case:

$$\begin{array}{ccc}
 \text{pairing of } N_\infty\text{-operads} & \begin{array}{c} \xrightarrow{\hspace{1cm}} \\ \xleftarrow[\text{partially}]{\hspace{1cm}} \end{array} & \text{compatible pair of transfer systems}
 \end{array}$$

Theorem 5 ([CCM⁺25]). *A pairing of N_∞ -operads determines a compatible pair of transfer systems. Conversely, if the additive transfer system is complete, then the compatible pair can be realized by a pairing of N_∞ -operads.*

This gives a homotopical realization of a combinatorial compatibility condition. We conjecture that the converse holds without this additional assumption and prove several further families of realizable pairs.

Operations are only one part of the story. Equivariant gradings themselves can carry additional structure. An ordinary $RO(C_2)$ -grading remembers only the class $[V]$ of a representation. Angeltveit and Bohmann introduced the category $\mathcal{RO}(G)$, which refines this grading by retaining morphisms between representation degrees, and in particular the symmetries of a representation V [AB18]. The distinction can be pictured at the level of graded Mackey functors:

$$\begin{array}{ccc}
 \text{\textit{RO}(C_2)\text{-grading}} & & \text{\textit{RO}(C_2)\text{-grading}} \\
 \begin{array}{c} \underline{M}_{[V]}(C_2/C_2) \\ \text{res} \downarrow \uparrow \text{tr} \\ \underline{M}_{[V]}(C_2/e) \end{array} & & \begin{array}{c} \underline{M}_V(C_2/C_2) \\ \text{res} \downarrow \uparrow \text{tr} \\ \underline{M}_V(C_2/e) \end{array} \quad \begin{array}{c} \curvearrowright \\ \text{Aut}(V) \end{array}
 \end{array}$$

Figure 5: $\mathcal{RO}(C_2)$ -grading retains the symmetries of the representation degree V .

In ongoing work with Noah Wisdom, we make this additional structure explicit. We study how $\mathcal{RO}(C_2)$ -graded Mackey functors can be described in terms of ordinary $RO(C_2)$ -graded data together with the

symmetries of the grading, and how this description extends to graded Tambara functors. The goal is to understand how these grading symmetries interact with equivariant operations, providing an algebraic framework for representation-graded coefficient systems arising in genuinely equivariant theories such as Real algebraic K -theory.

Together, these projects study two basic features of genuine equivariant invariants: the operations they support and the structure carried by their gradings. Both appear naturally in Real algebraic K -theory and in my work on Real trace methods, where the relevant invariants are genuinely equivariant and representation-graded. My broader goal is to develop an algebraic language in which this structure is retained when passing to computations and, eventually, to equivariant forms of descent.

3 Current research and direction

Building on the framework of stable homotopy theory, my current research explores how algebraic and homotopical ideas interact across several fronts, particularly through *algebraic K-theory*, *equivariant algebra*, and their connections to number theory and combinatorics. Rather than treating these areas in isolation, I study how organizing principles such as duality, symmetry, and operadic compatibility arise in parallel forms across them. The following sections describe how these themes manifest in my recent and ongoing work.

4 Future research direction

My future research builds directly on my current work, extending results on duality phenomena in algebraic K-theory and advancing the foundations of equivariant algebra. A central aim is to further develop the connections between homotopical methods, arithmetic dualities, and symmetry in algebraic and geometric contexts.

In parallel, I plan to develop projects that emphasize concrete computations and examples, especially in areas such as homotopical combinatorics and topological data analysis. These topics are well suited for undergraduate involvement and provide natural entry points into algebraic topology. I view cultivating such accessible research directions as an essential part of advancing the field and training the next generation of mathematicians.

Foundations for Equivariant Algebra A key direction of my future work is to develop the basic algebraic framework underlying equivariant stable homotopy theory. While advanced constructions such as Tambara functors capture much of the structure, many of the elementary notions familiar from commutative algebra remain to be formulated in the equivariant setting.

A natural entry point is the study of commutative G -rings, a simpler type of equivariant algebraic object that Tambara functors generalize. Even in this case, elementary concepts such as algebraic extensions and algebraic closures exhibit subtle equivariant pathologies. Investigating these questions offers an accessible entry point that also exposes conceptual obstructions in the more general theory of Tambara functors. Beginning with these manageable cases allows one to develop the algebraic intuition and categorical techniques needed to approach the full equivariant framework.

My goal is to establish such foundational concepts and structures within the equivariant setting. This foundational program would provide the analogue of commutative algebra for equivariant algebraic geometry, bridging the gap between algebraic and homotopical formulations of equivariant descent.

Undergraduate Research Directions Several aspects of my work give rise to problems and projects that are accessible at an undergraduate level yet remain closely connected to broader research questions. Two areas in particular, combinatorial transfer systems and topological data analysis (TDA), offer tractable problems that both engage students and contribute to the development of my research programs.

One direction of my work focuses on the study of *transfer systems* from a combinatorial and educational perspective. Because transfer systems are defined directly in terms of subgroup lattices, they are easy to state and visualize, making them particularly suitable for undergraduate research projects. At the same time, their enumeration already reveals rich combinatorial structure: for example, the number of transfer systems for cyclic groups coincides with the *Catalan numbers* [BBR21], and many related patterns remain to be understood. Exploring these systems provides an accessible way to connect discrete mathematics with deeper questions in equivariant topology. Counting and classifying finite posets associated with transfer systems not only offer combinatorial insight but also inform the structural classification of N_∞ -operads and Tambara functors, clarifying how discrete data govern admissible algebraic operations. In this way, the topic

serves both as an entry point for students and as a bridge between combinatorics and the homotopical study of equivariant algebra.

Another area is *topological data analysis* (TDA), which applies algebraic topology to study the structure of high-dimensional data sets. Methods such as *persistent homology* can be introduced in a visual and computationally accessible way, yet they lead naturally into deeper questions about stability, functoriality, and the role of cohomology operations. These connections remain an active area of research, and concrete case studies in TDA provide a natural testing ground for exploring how homotopical invariants can yield new insights into the shape and geometry of data.

In this way, projects in combinatorics and data science serve not only as entry points for undergraduate participation but also as stepping stones toward more advanced problems aligned with my core research agenda.

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